

ULTIMATE GCSE CHEMISTRY REVISION GUIDE

Full booklet covering all 3 topics from your revision list

Chemical Reactions | Atomic Structure | Periodic Table, Metals & Materials



Definitions | diagrams | equations | worked examples | exam traps

Designed for clear, high-quality GCSE revision

How to use this booklet

This guide covers the full chemistry revision list you gave: chemical reactions, atomic structure, and the periodic table with metals, non-metals, alloys and metal reactions. It is written for strong GCSE-style answers: precise definitions, particle explanations, equations, observations, worked examples and common exam traps.

How to score higher

For exam questions, do not only name facts. Explain why: use atoms, bonds, electrons, energy transfer, shielding and conservation of mass.

Contents

1. Chemical and physical changes
2. Reactants, products and conservation of mass
3. Equations and balancing
4. Exothermic and endothermic reactions
5. Combustion, oxidation and reduction
6. Thermal decomposition and neutralisation
7. Catalysts and practical observations
8. Atoms, subatomic particles and electron shells
9. Ions, isotopes, relative atomic mass and carbon dating
10. Periodic table, Mendeleev, elements, compounds and mixtures
11. Metals, non-metals, alloys, reactions with water/acids, and trends
12. Practice questions and answers

TOPIC 1

CHEMICAL REACTIONS

Physical change, chemical change, equations, energy, combustion, oxidation, neutralisation and catalysts



REVISION SECTION

1. Chemical and Physical Changes

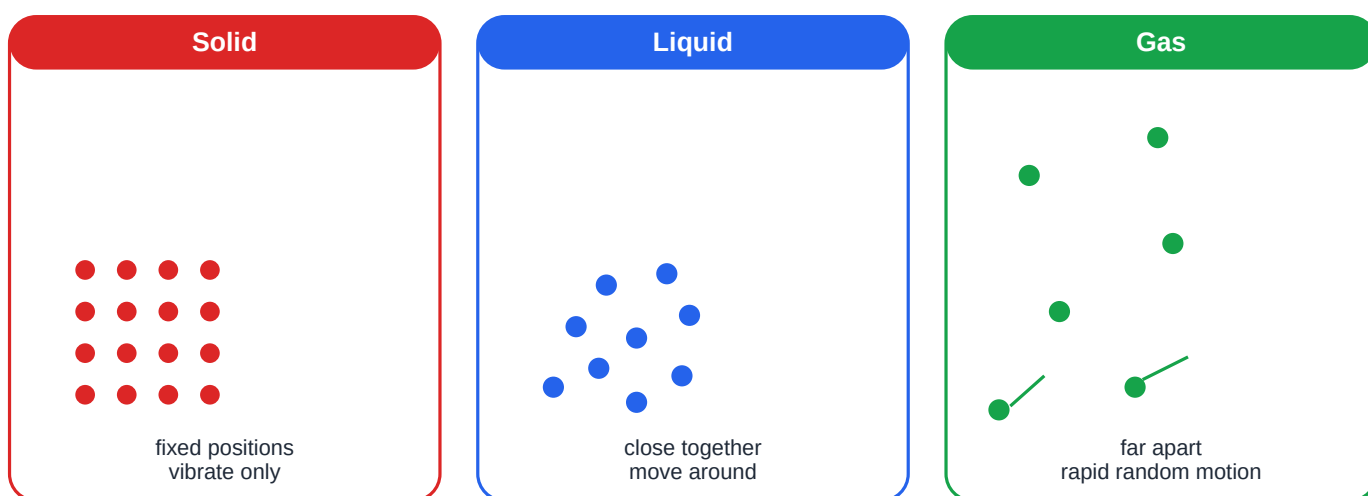
Chemistry begins with deciding whether a change is physical or chemical. The most important question is: **has a new substance been formed?** If no new substance forms, it is a physical change. If new products form with different properties, it is a chemical change.

Physical changes

A physical change alters the appearance, shape, size, state or arrangement of a substance without changing its chemical identity. The same substance is present before and after the change.

- No new substance is made.
- Usually reversible, especially changes of state.
- Particles stay the same substance.
- No chemical bonds are broken or formed.
- Only particle arrangement, movement or energy changes.

Examples include melting ice, boiling water, freezing water, condensation, sublimation, dissolving sugar and cutting paper.



Precision point

Do not call these 'physical reactions'. The better scientific term is **physical changes** because no chemical reaction has taken place.

Chemical changes

A chemical change produces one or more new substances. The starting substances are chemically changed into products with different properties.

- Existing chemical bonds break.
- Atoms rearrange into new combinations.
- New bonds form.
- New products are made.
- Often difficult to reverse, but not always impossible. Some chemical reactions are reversible.

Examples include combustion, rusting, oxidation, neutralisation, thermal decomposition, cooking an egg, burning wood, and vinegar reacting with baking soda.

Grade 8-9 wording

A top answer says: 'In a chemical reaction, atoms are rearranged. Bonds are broken and new bonds are formed. New substances are produced, but atoms are not created or destroyed.'

2. Evidence of Chemical Reactions

You often cannot see individual atoms or bonds, so chemists use observations as evidence that a chemical reaction has occurred.

Observation	What it suggests	Exam-safe wording
Fizzing / bubbling / effervescence	A gas is produced	Gas is produced, causing fizzing.
Precipitate	An insoluble solid has formed	A solid precipitate forms and the solution may turn cloudy.
Temperature rise	Energy transferred to surroundings	The reaction is exothermic.
Temperature fall	Energy taken from surroundings	The reaction is endothermic.
Colour change	New substances may have formed	A colour change is observed.
Light / flame	Energy released	Light is emitted during the reaction.

Common mistake

Never write 'I would see oxygen' or 'I would see hydrogen'. You cannot identify a gas just by looking. Write 'bubbles/fizzing/gas produced' unless a test for the gas is described.

3. Reactants and Products

Chemical equations show what substances react and what substances are made.

reactants -> products

Reactants are on the left; products are on the right

Reactants are the starting substances present before the reaction. Products are the new substances formed by the reaction.

magnesium + oxygen -> magnesium oxide

Reactants: magnesium and oxygen. Product: magnesium oxide.

In symbol form this is written:



Balanced symbol equation for magnesium burning in oxygen

4. Conservation of Mass

The law of conservation of mass states that atoms cannot be created or destroyed in a chemical reaction. They are only rearranged into new products. This means the total mass of reactants equals the total mass of products.

mass of reactants = mass of products

Because atoms are conserved

Example: if 0.20 g of zinc reacts with 0.36 g of oxygen, the mass of zinc oxide formed is 0.56 g.

0.20 g + 0.36 g = 0.56 g

zinc + oxygen → zinc oxide

Exam trap

If a gas escapes, the measured mass may appear to decrease. This does not break conservation of mass; it means some product has left the container.

5. Chemical Equations and Balancing

Word equations use names of substances. Symbol equations use chemical formulae. Symbol equations must be balanced because every atom on the reactant side must still exist on the product side.

Balancing rules:

- Count atoms of each element on both sides.
- Add big numbers in front of formulae called coefficients.
- Never change the small numbers inside a formula, because that changes the substance.
- Recount and check every atom type.

H₂ + O₂ → H₂O

Unbalanced: oxygen atoms do not match

2H₂ + O₂ → 2H₂O

Balanced: 4 H atoms and 2 O atoms on both sides

Unbalanced	Balanced	Reaction type
Mg + O ₂ → MgO	2Mg + O ₂ → 2MgO	oxidation / combustion of metal
H ₂ + Cl ₂ → HCl	H ₂ + Cl ₂ → 2HCl	combination reaction
CH ₄ + O ₂ → CO ₂ + H ₂ O	CH ₄ + 2O ₂ → CO ₂ + 2H ₂ O	complete combustion

6. Exothermic and Endothermic Reactions

Chemical reactions involve energy changes because energy is needed to break bonds and energy is released when new bonds form.

Exothermic reactions

An exothermic reaction transfers energy to the surroundings. The surroundings become warmer and the measured temperature increases.

- Energy is released overall.
- Temperature rises.
- Examples: combustion, respiration, neutralisation and fireworks.

Endothermic reactions

An endothermic reaction takes in energy from the surroundings. The surroundings lose thermal energy, so the temperature decreases.

- Energy is absorbed overall.
- Temperature falls.
- Examples: photosynthesis, thermal decomposition and some instant cold packs.

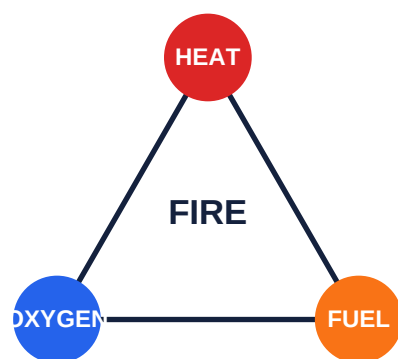
Reaction type	Energy direction	Temperature change	Examples
Exothermic	to surroundings	increases	combustion, neutralisation, respiration
Endothermic	from surroundings	decreases	thermal decomposition, photosynthesis, cold packs

Memorise this

Best wording: 'Exothermic reactions transfer energy to the surroundings. Endothermic reactions take in energy from the surroundings.'

7. Combustion

Combustion is burning in oxygen. It is a rapid reaction between a fuel and oxygen. Combustion is an exothermic reaction because it releases energy to the surroundings as heat and often light.



Remove one side and combustion stops

The three things needed for combustion are fuel, oxygen and heat. This is called the fire triangle. Remove any one of these and the fire stops.

Complete combustion of hydrocarbons

A hydrocarbon is a compound made of hydrogen and carbon. In complete combustion, there is enough oxygen for the fuel to burn fully.

hydrocarbon + oxygen -> carbon dioxide + water + energy

General equation for complete combustion

CH₄ + 2O₂ -> CO₂ + 2H₂O

Complete combustion of methane

Exam link

Complete combustion of hydrocarbons always produces carbon dioxide and water. If carbon monoxide or soot forms, that is incomplete combustion, which may not be in your test but is useful context.

8. Oxidation and Reduction

Oxidation can be described in two ways. At your level, both are useful.

Term	Oxygen definition	Electron definition
Oxidation	gain of oxygen	loss of electrons
Reduction	loss of oxygen	gain of electrons

The memory aid is **OIL RIG**: Oxidation Is Loss, Reduction Is Gain - of electrons.



Iron is oxidised because it gains oxygen

Combustion is a type of oxidation because the fuel reacts with oxygen. All combustion involves oxidation, but not every oxidation reaction is combustion.

Model answer

If asked why combustion is oxidation, write: 'Combustion involves a substance reacting with oxygen. Oxidation is the gain of oxygen, so the fuel is oxidised.'

9. Thermal Decomposition

Thermal decomposition is a chemical reaction in which a compound breaks down into simpler substances when heated. It is usually endothermic because heat energy is absorbed to break bonds in the compound.



General pattern for metal carbonates



Example of thermal decomposition

This reaction produces carbon dioxide gas. If tested, carbon dioxide turns limewater cloudy.

Common mistake

Do not use combustion as an example of thermal decomposition. Combustion is burning in oxygen; thermal decomposition is breaking a compound down using heat.

10. Neutralisation

Neutralisation is a reaction between an acid and an alkali. The products are a salt and water.



General neutralisation equation



Specific example

The ionic explanation is that hydrogen ions from the acid react with hydroxide ions from the alkali to form water.



Net ionic equation

Neutralisation is usually exothermic, so the temperature often increases.

11. Catalysts and Speeding Up Reactions

A catalyst is a substance that speeds up a chemical reaction without being used up or permanently changed. It can be reused because it comes out of the reaction unchanged.

- Catalysts lower activation energy.
- Activation energy is the minimum energy particles need to react.
- Lower activation energy means more collisions are successful.
- Catalysts do not change the amount of product made; they affect speed.



Hydrogen peroxide decomposes into water and oxygen

Manganese dioxide is a common catalyst for the decomposition of hydrogen peroxide. It speeds up oxygen production but is not used up.

Other ways to speed reactions

Method	Why rate increases
Increase temperature	Particles move faster and collide more often with more energy.
Increase concentration	More particles in the same volume means more frequent collisions.
Increase surface area	More particles are exposed to collide with reactants.
Add a catalyst	Activation energy is lowered, so more collisions succeed.

TOPIC 2

ATOMIC STRUCTURE

Atoms, subatomic particles, mass number, electron shells, ions, isotopes,
relative atomic mass and carbon dating



REVISION SECTION

12. What Matter Is Made Of

All matter is made from tiny particles called atoms. Atoms are the smallest particles of an element that still have the chemical properties of that element.

Different substances are made because atoms can be arranged and bonded in different ways. Chemistry studies how atoms are structured and how they rearrange in reactions.

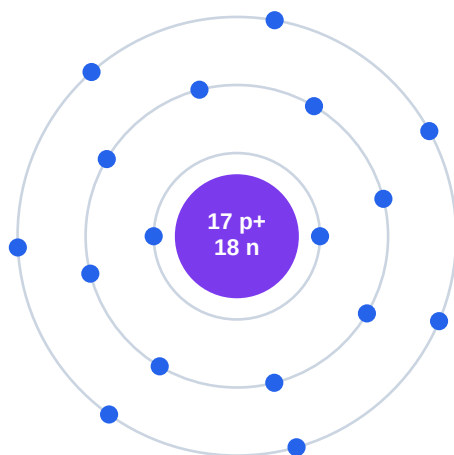
Atoms, molecules and structures

Word	Meaning	Example
Atom	Smallest particle of an element	one oxygen atom, O
Molecule	Two or more atoms chemically bonded	O ₂ , H ₂ O, CO ₂
Structure	A larger arrangement or pattern of atoms/particles	metal lattice, giant structure, crystal
Element	Made from one type of atom only	iron, oxygen, carbon
Compound	Two or more different elements chemically bonded	water, sodium chloride
Mixture	Substances physically combined, not chemically bonded	air, salt water

13. Basic Structure of the Atom

An atom has a small central nucleus containing protons and neutrons. Electrons are arranged in shells around the nucleus. Most of the atom is empty space.

Chlorine atom: electron arrangement 2,8,7



Protons and neutrons in nucleus; electrons in shells

Particle	Relative charge	Relative mass	Location
Proton	+1	1	nucleus
Neutron	0	1	nucleus
Electron	-1	very small	electron shells

Key point

In a neutral atom, the number of protons equals the number of electrons. Neutrons do not affect charge.

14. Atomic Number, Mass Number and Neutrons

The atomic number identifies the element because it tells you the number of protons. If the number of protons changes, it becomes a different element.

atomic number = number of protons

This identifies the element

mass number = protons + neutrons

Electrons are not counted because their mass is tiny

neutrons = mass number - atomic number

Use this for calculations

Atom	Atomic number	Mass number	Protons	Neutrons	Electrons in neutral atom
Carbon-12	6	12	6	6	6
Sodium-23	11	23	11	12	11
Chlorine-35	17	35	17	18	17
Potassium-39	19	39	19	20	19

15. Electron Configuration

Electrons occupy shells around the nucleus. For the first 20 elements, the simplified shell filling is 2, then 8, then 8, then 2.

1st shell = 2 2nd shell = 8 3rd shell = 8

Simplified GCSE/KS3 shell model

Element	Atomic number	Electron arrangement	Outer electrons	Periodic table link
Oxygen	8	2,6	6	Group 6, Period 2
Sodium	11	2,8,1	1	Group 1, Period 3
Magnesium	12	2,8,2	2	Group 2, Period 3
Chlorine	17	2,8,7	7	Group 7, Period 3
Calcium	20	2,8,8,2	2	Group 2, Period 4

Periodic table link

Group number tells you the number of outer-shell electrons for main-group elements. Period number tells you the number of occupied shells.

16. Ions

Atoms form ions by gaining or losing electrons. This changes the charge because electrons are negative. The nucleus does not change in normal chemical reactions.

Positive ions

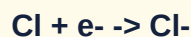
Metals lose electrons to form positive ions. They become positive because they now have more protons than electrons.



Sodium loses one electron: 2,8,1 → 2,8

Negative ions

Non-metals gain electrons to form negative ions. They become negative because they now have more electrons than protons.



Chlorine gains one electron: 2,8,7 → 2,8,8

Atom	Starts as	Ion formed	What happens	Final arrangement
Sodium	2,8,1	Na ⁺	loses 1 electron	2,8
Magnesium	2,8,2	Mg ²⁺	loses 2 electrons	2,8
Chlorine	2,8,7	Cl ⁻	gains 1 electron	2,8,8
Oxygen	2,6	O ²⁻	gains 2 electrons	2,8

17. Isotopes

Isotopes are atoms of the same element with the same number of protons but different numbers of neutrons. Because the proton number is the same, they are the same element. Because neutron number differs, their mass number differs.

same protons + different neutrons = isotopes

Same element, different mass number

Isotope	Protons	Neutrons	Mass number
Carbon-12	6	6	12
Carbon-14	6	8	14
Chlorine-35	17	18	35
Chlorine-37	17	20	37

Isotopes have the same chemical properties because chemical reactions depend mainly on electrons, and isotopes of the same element have the same electron arrangement. However, they can have different physical properties because their masses are different. Some isotopes are radioactive, meaning their nuclei are unstable.

High-mark sentence

If asked why isotopes react similarly: 'They have the same electron arrangement, so they have the same chemical properties.'

18. Relative Atomic Mass

Relative atomic mass, A_r , is the weighted average mass of an element's naturally occurring isotopes. It is weighted because some isotopes are more common than others.

$A_r = \text{sum of (isotope mass} \times \text{abundance)} / \text{total abundance}$

Use abundances as percentages or ratios

Relative atomic mass worked example: chlorine

Chlorine-35 = 75% abundance

Chlorine-37 = 25% abundance

$A_r = [(35 \times 75) + (37 \times 25)] / 100$

$A_r = (2625 + 925) / 100 = 35.5$

This explains why chlorine has a relative atomic mass of about 35.5 rather than a whole number. Most chlorine atoms are chlorine-35, but a smaller number are chlorine-37.

Exam trap

Do not just average isotope masses unless they have equal abundance. Always account for how common each isotope is.

19. Carbon Dating

Carbon dating uses the radioactive isotope carbon-14 to estimate the age of once-living materials. Living organisms take in carbon while alive. After death, they stop taking in carbon and the amount of carbon-14 gradually decreases through radioactive decay.

- Used for fossils, bones, ancient wood and archaeological remains.
- Works only for once-living materials, not ordinary rocks or metals.
- The less carbon-14 remains, the older the sample is.

Scope

You do not need the full radioactive decay mathematics unless your teacher gives it, but you should know the purpose and principle of carbon dating.

TOPIC 3

PERIODIC TABLE, METALS & MATERIALS

Mendeleev, elements, compounds, mixtures, groups, periods, metals,
non-metals, alloys and reactivity



REVISION SECTION

20. Development of the Periodic Table

The modern periodic table arranges elements in order of increasing atomic number. It also groups elements with similar chemical properties together.

Dmitri Mendeleev

Dmitri Mendeleev was a Russian chemist who helped develop the periodic table. He arranged elements mainly by atomic mass and chemical properties. His major achievement was that he left gaps for undiscovered elements and predicted their properties.

- He did not force every element into the wrong place just to follow mass order.
- He used patterns in properties to improve the arrangement.
- Later discoveries matched many of his predictions, supporting his table.

Why it matters

Mendeleev is important because he used evidence and patterns to predict unknown elements. This is a strong example of scientific modelling.

21. Elements, Compounds and Mixtures

You need to clearly separate elements, compounds and mixtures because exam questions often test this using diagrams or formulas.

Type	Definition	Example	How separated
Element	One type of atom only	oxygen, iron, gold, carbon	cannot be chemically broken into simpler substances
Compound	Two or more different elements chemically bonded	H ₂ O, CO ₂ , NaCl	chemical reactions
Mixture	Substances physically combined, not bonded	air, salt water	physical methods such as filtration, evaporation, chromatography

A compound has properties different from the elements that form it. For example, sodium is a reactive metal and chlorine is a toxic gas, but sodium chloride is table salt.

22. Periods and Groups

The periodic table is organised so that electron arrangement links directly to position.

period number = number of occupied electron shells

Rows go across

group number = number of outer-shell electrons

Columns go down

Example: sodium has electron arrangement 2,8,1. It has 3 occupied shells, so it is in Period 3. It has 1 outer electron, so it is in Group 1.

Example: chlorine has electron arrangement 2,8,7. It has 3 shells, so it is in Period 3. It has 7 outer electrons, so it is in Group 7.

Group 1

reactivity increases down



outer electron lost more easily

Group 7

reactivity decreases down



incoming electron attracted less strongly

23. Group 1 - Alkali Metals

Group 1 elements include lithium, sodium and potassium. They are reactive metals and have one electron in their outer shell. They react by losing this one electron to form +1 ions.

- Soft metals.
- Low density compared with many other metals.
- React with water to form metal hydroxide and hydrogen.
- Reactivity increases down the group.



General equation for Group 1 with water



Example

Why Group 1 reactivity increases

Down the group, atoms have more electron shells. The outer electron is further from the nucleus and more shielded by inner shells. The attraction between the nucleus and the outer electron is weaker, so the electron is lost more easily. Therefore reactivity increases down Group 1.

Memory line

Group 1 = loses electrons. Down group = easier to lose. Therefore reactivity increases.

24. Group 7 - Halogens

Group 7 elements include fluorine, chlorine, bromine and iodine. They are reactive non-metals and have seven electrons in the outer shell. They react by gaining one electron to form -1 ions.

- Diatomic molecules, such as Cl₂ and Br₂.
- Form -1 ions.
- Reactivity decreases down the group.
- More reactive halogens can displace less reactive halogens from compounds.

Why Group 7 reactivity decreases

Down the group, atoms have more electron shells and greater shielding. The incoming electron is attracted less strongly by the nucleus because it is further away and shielded by inner shells. It becomes harder to gain an electron, so reactivity decreases down Group 7.

Memory line

Group 7 = gains electrons. Down group = harder to gain. Therefore reactivity decreases.

25. Group 0 - Noble Gases

Group 0 elements include helium, neon and argon. They are very unreactive because their outer electron shells are full. This makes them stable, so they do not need to gain, lose or share electrons.

- Full outer shell.
- Very unreactive/inert.
- Colourless gases at room temperature.
- Exist as single atoms rather than molecules.

Key explanation

Noble gases are stable because their outer shell is already full. This is the reason they rarely form bonds.

26. Metals and Non-metals

Metals and non-metals have different physical and chemical properties. Their properties are linked to how their particles and electrons behave.

Metals	Non-metals
Usually shiny/lustrous	Usually dull
Good conductors of heat and electricity	Poor conductors, with exceptions such as graphite
Malleable - can be hammered into shape	Brittle when solid
Ductile - can be drawn into wires	Not usually ductile
Usually high melting points	Often lower melting points
Usually form positive ions	Usually form negative ions or share electrons

Metals are found mainly on the left and centre of the periodic table. Non-metals are found mainly on the right.

Metalloids / semi-metals

Metalloids, also called semi-metals, have properties between metals and non-metals. Examples include silicon and boron. They can be useful in electronics because their conductivity is between that of metals and non-metals.

27. Alloys

An alloy is a mixture containing a metal and one or more other elements. Alloys are often used instead of pure metals because their properties are improved.

Alloy	Main substances	Useful properties / uses
Steel	iron + carbon	stronger and harder than pure iron; buildings, tools, vehicles
Stainless steel	iron + chromium + sometimes nickel	resists corrosion; cutlery, medical equipment
Brass	copper + zinc	harder than copper; instruments, fittings, coins
Bronze	copper + tin	hard, corrosion resistant; statues, medals

Pure metals can be soft because layers of atoms can slide over each other. In an alloy, atoms of different sizes distort the layers, making it harder for them to slide. This usually makes the alloy harder and stronger.

Specification link

Your teacher's 'alloys and what they are made of' point means you should know examples like steel = iron + carbon and brass = copper + zinc.

28. Reactions of Metals with Water and Acids

Metals with water

Some metals react with water to form a metal hydroxide and hydrogen gas.



General equation



Example

Common observations include fizzing, movement on the water surface, heat production and hydrogen gas release.

Metals with acids

Many metals react with acids to produce a salt and hydrogen gas.



General equation



Example

Common observations include bubbles/fizzing, the metal disappearing, and a temperature increase if the reaction is exothermic.

Observation wording

If a gas is produced in these metal reactions, the safe observation is 'fizzing/bubbles/effervescence', not 'I can see hydrogen'.

29. Reactivity Series and Corrosion

The reactivity series places metals in order of how easily they react. More reactive metals lose electrons more easily. Metals such as potassium and sodium are very reactive; metals such as copper, silver and gold are much less reactive.

Corrosion is the gradual destruction of a material by chemical reactions with the environment. Rusting is corrosion of iron. Iron reacts with oxygen and water to form hydrated iron oxide, commonly called rust.



Simplified rusting equation

Alloys are often chosen because they resist corrosion better than pure metals. Stainless steel contains chromium, which helps protect the metal from rusting.

Composites, polymers and ceramics

Your revision list mentioned materials such as composites, polymers and ceramics. These are broad material classes often compared with metals.

Material class	Meaning	Examples / properties
Polymer	Long-chain molecules made from repeating units	plastics; often light and flexible
Ceramic	Hard, heat-resistant non-metal material	glass, pottery; often brittle
Composite	Material made from two or more materials combined	fibreglass, reinforced concrete; combines useful properties

30. Ultra-Important Definitions

Term	Definition
Physical change	A change in state, shape or appearance without forming a new substance.
Chemical change	A change that forms new substances through bond breaking and bond forming.
Reactant	Starting substance in a chemical reaction.
Product	Substance formed in a chemical reaction.
Catalyst	Substance that speeds up a reaction without being used up.
Atom	Smallest particle of an element that keeps that element's chemical properties.
Element	Pure substance made from one type of atom.
Compound	Substance made from two or more different elements chemically bonded.
Mixture	Substances physically combined without chemical bonding.
Molecule	Two or more atoms chemically bonded together.
Isotope	Atoms of the same element with different numbers of neutrons.
Alloy	Mixture containing a metal and one or more other elements.
Exothermic	Transfers energy to surroundings.
Endothermic	Takes in energy from surroundings.
Oxidation	Gain of oxygen or loss of electrons.
Reduction	Loss of oxygen or gain of electrons.

31. Exam Sentence Bank

Use these sentence patterns when answering longer questions.

- A chemical change forms new substances because bonds are broken and new bonds are formed.
- A physical change does not form a new substance; only the arrangement or energy of particles changes.
- The equation must be balanced because atoms cannot be created or destroyed.
- The total mass of reactants equals the total mass of products because atoms are conserved.
- Combustion is oxidation because the fuel reacts with oxygen and gains oxygen.
- A catalyst lowers activation energy and is not used up, so it can be reused.
- Group 1 reactivity increases down the group because the outer electron is further from the nucleus and more shielded, so it is lost more easily.
- Group 7 reactivity decreases down the group because the incoming electron is less strongly attracted to the nucleus due to increased shielding and distance.
- Isotopes have the same chemical properties because they have the same electron arrangement.
- Relative atomic mass is a weighted average because isotopes have different abundances.

32. Practice Questions - No Multiple Choice

Answer these in full sentences. These cover the full booklet.

No.	Question
1	Explain the difference between a chemical change and a physical change using particles and bonds.
2	State three observations that suggest a chemical reaction has occurred, and explain why 'seeing oxygen' is poor wording.
3	Define reactants and products using magnesium + oxygen → magnesium oxide.
4	Balance: $\text{H}_2 + \text{O}_2 \rightarrow \text{H}_2\text{O}$ and explain why balancing is required.
5	Explain conservation of mass using atoms.
6	Compare exothermic and endothermic reactions using energy transfer and temperature change.
7	Write the general equation for complete combustion of a hydrocarbon and explain why combustion is oxidation.
8	Define thermal decomposition and give the general equation for a metal carbonate.
9	Explain neutralisation using acid, alkali, salt, water, H^+ and OH^- .
10	Define catalyst and explain activation energy.
11	For chlorine-35, find protons, neutrons and electrons.
12	Explain what happens when sodium forms Na^+ .
13	Define isotope and explain why isotopes react similarly.
14	Calculate A_r for 80% X-10 and 20% X-11.
15	Explain why Group 1 increases and Group 7 decreases in reactivity down the group.
16	Compare metals, non-metals and alloys.
17	Write equations for metal + water and metal + acid.
18	Explain Mendeleev's importance in the periodic table.

33. Practice Answers - Condensed Mark Scheme

No.	Key answer points
1	Physical: no new substance, usually reversible, no bonds broken/formed. Chemical: new substance, atoms rearranged, bonds break/form.
2	Colour change, temperature change, gas/fizzing, precipitate, light, smell. You observe gas production, not the gas identity by sight.
3	Reactants are magnesium and oxygen. Product is magnesium oxide.
4	$2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$. Atoms must match because atoms cannot be created or destroyed.
5	Mass conserved because the same atoms are present before and after, only rearranged.
6	Exothermic transfers energy to surroundings, temperature rises. Endothermic absorbs energy, temperature falls.
7	Hydrocarbon + oxygen $\rightarrow$ carbon dioxide + water + energy. Oxidation because fuel gains oxygen/reacts with oxygen.
8	A compound breaks down using heat. Metal carbonate $\rightarrow$ metal oxide + carbon dioxide.
9	Acid + alkali $\rightarrow$ salt + water. $\text{H}^+ + \text{OH}^- \rightarrow \text{H}_2\text{O}$.
10	Speeds reaction without being used up. Lowers activation energy.
11	17 protons, 18 neutrons, 17 electrons for neutral chlorine-35.
12	Sodium loses one electron, forms Na^+ , arrangement 2,8, stable full shell.
13	Same protons, different neutrons. Same electron arrangement means similar chemical properties.
14	$(10 \times 80 + 11 \times 20)/100 = 10.2$.
15	Group 1 loses electrons more easily down group. Group 7 gains electrons less easily down group due to shielding/distance.
16	Metals conduct and are malleable. Non-metals poor conductors/brittle. Alloys are mixtures containing metals with improved properties.
17	Metal + water $\rightarrow$ metal hydroxide + hydrogen. Metal + acid $\rightarrow$ salt + hydrogen.
18	Arranged by patterns, left gaps, predicted missing elements.

34. Final Checklist

You are ready when you can do all of these without notes.

- Explain chemical vs physical changes using particles, bonds and products.
- Identify observations: fizzing, temperature change, colour change, precipitate, light.
- Identify reactants and products in equations.
- Balance simple chemical equations.
- Explain conservation of mass.
- Define exothermic and endothermic and give examples.
- Explain combustion using the fire triangle and hydrocarbon products.
- Define oxidation and reduction.
- Explain thermal decomposition, neutralisation and catalysts.
- Calculate protons, neutrons and electrons from atomic and mass numbers.
- Draw electron arrangements using shells.
- Explain ions and full outer shells.
- Define isotopes and calculate relative atomic mass.
- Explain carbon dating.
- Explain Mendeleev, groups, periods and electron arrangement.
- Compare metals, non-metals, metalloids, alloys, polymers, ceramics and composites.
- Write word equations for metals reacting with water and acids.
- Explain Group 1 and Group 7 trends using shielding and distance from the nucleus.

Coverage note

This booklet deliberately includes all content from your long revision list and the extra teacher points: physical/chemical changes, observations, reactants/products, equations, exothermic/endothermic, combustion, oxidation/reduction, thermal decomposition, neutralisation, catalysts, atomic structure, isotopes, relative atomic mass, carbon dating, Mendeleev, elements/compounds/mixtures, metals/non-metals, alloys, materials and metal reactions.